Mathematical overview

Start R1,B1 and choose uniformly among current counter identities, returning every draw. In the usual copying rule, two red-draw totals 0,1,2 have probabilities 2/6,2/6,2/6. If each returned draw instead adds one counter of the OTHER color, those totals have probabilities 1/6,4/6,1/6. With no addition they have probabilities 1/4,2/4,1/4. In this two-draw comparison, opposite reinforcement promotes switching and balanced draw totals; copying promotes repeating. No general balance law follows from this one comparison. The six add-a-counter histories each have probability (1/2)(1/3), while the four no-add histories each have (1/2)(1/2). Equal histories must be compared within their own mechanism, then via ratios; the different raw denominators cannot be ignored.

Under copying, after a history with r red draws and b blue draws, the current bag is (r+1)R,(b+1)B. The next red chance is (r+1)/(r+b+2), not the original 1/2. The order does not affect this forecast because the count bag is the same. After four draws the red chance is 1/6,2/6,3/6,4/6 or 5/6. Extremal design is attainable by BBBB or RRRR; a tied forecast requires two of each color. Different orders with the same totals give the same next-draw prediction. This is dependence across draws, not evidence for independent trials with a fixed p.

Start one counter of each of three colors and copy the drawn color. After n draws the added-count vector (r,b,g) ranges over all nonnegative triples summing to n. Each vector has chance 1/C(n+2,2). Each particular word has probability 2r!b!g!/(n+2)!, so words need not be equally likely. There are (n+2)!/2 equal marked-identity histories and n! histories per vector. For n=2: twelve histories, six vectors, two each. For n=3: sixty histories, ten vectors, six each. The final bag counts are (r+1,b+1,g+1); original colors never vanish. The theorem assumes one initial counter per color, one same-color addition per returned draw and uniform identity selection. Other initial bags or feedback rules can give different laws.

Grades 2-3 can compare short feedback rules and make concrete next-draw bags. Grades 4-5 can sort three-color histories, explain equal vector weights and extend the count law. K-1 can enact the opposite-color addition and build strongest/weakest/tied bags with adult reading and counting to six; the complete marked probability comparison remains a readiness-dependent continuation. Experiments produce conjectures; exact marked histories and count-product arguments establish the chance claims.

Materials, route, and flexible time

Per pair: one opaque bag/cup, six red, six blue and six green same-size/same-feel counters, a tray for the outside supply, removable labels 1-5, pencil and three student pages. Only the stated originals start inside; reset before every new story. Labels name color plus birth step, as in the base packet: original R1/B1/G1; the first new copy is R2, B2 or G2 depending on its color, the next is 3. They do not number same-color counters consecutively. A second page-3 print can be cut into the twelve supplied movable history cards by an adult; children may write the final color counts on those cards and pool/sort them. Cutting stays in preparation, not a child task. For the three-draw return, add ten blank 100-by-60 mm count-record cards with three columns labeled R, B, G, thirty blank 70-by-25 mm history slips, and two blank Letter sheets per pair. These are paper records rather than physical counter-fit mats: children retain one card for each final count bag and place supporting words/count evidence beneath it. If choosing full identity enumeration rather than color-word weights, supply thirty further slips on that return visit. There is no exact-size board fit claim.

For eleven children at K–1 / 2–3 / 4–5: five kits, five bags/trays, thirty counters of each color (90 total), five label sets and pencils, fifty count-record cards, 150 blank history slips and ten blank Letter sheets (150 further slips only if all five kits choose full three-draw identity enumeration). The upper third child rotates as banker/checker. Print five shared companions and five extra copies of page 3 for movable cards, or eleven selected recording pages as desired. Three anchored adults supervise the addition rule and resets while children choose stories, construct bags and compare forecasts. A partner handles the outside supply and checks each returned counter plus exactly one legal addition.

Shared launch, 4-6 minutes: let children handle R/B. Demonstrate the non-target single opposite-reinforcement step R1,B1 -> draw R1 -> return R1 and add B2 -> R1,B1,B2, as pictured. Let

a child select a legal next identity. Explicitly restore originals before changing rules. Before Problem 3, demonstrate G1 returning and adding G2 in R1,B1,G1; the next draw can now choose G2. Distinguish deliberately chosen possible stories from hidden random draws.

A manageable first visit can spend 15-25 minutes comparing two rules, then 15-20 on forecasts. Add the no-add comparison only after children know which bag the second draw uses. Three-color copying can be a separate 20-35 minute visit. Pooled twelve-card sorting is enough before extending to sixty histories; no one should transcribe a sixty-row table. Before a three-draw return, place the ten blank count cards and history slips in the kit. Let children choose predicted count bags and retain them on separate cards, then attach exact evidence; cards with the same final R/B/G counts can be merged. Children may extend the twelve printed two-draw cards by listing the five possible third identities, pooling first-draw groups and using one recoverable story per slip. Alternatively, retain each distinct color word on one of 27 slips, use actual available same-color choices to find its identity multiplicity, and attach that weight to its final-bag card. A child need not duplicate either complete collection. Do not silently operate the whole investigation for them: offer a choice of these record routes only when needed. The oldest children may use count-group transitions or a reversible count argument instead. Finishing every page is unnecessary.

Problem 1 (page 1)

For copying the complete identity histories are (R1,R1),(R1,R2),(R1,B1),(B1,R1),(B1,B1),(B1,B2). The middle two give one red and one blue draw, hence 2/6=1/3.

For opposite reinforcement: (R1,R1) has two red draws; (R1,B1),(R1,B2),(B1,R1),(B1,R2) have one red; (B1,B1) has zero red. These six are equally likely. The no-add column has two spare lines; the six printed lines in a column do not assert six legal histories under every rule. Thus one of each has chance 4/6=2/3, greatest among the three rules. Note that added counter B2 exists after R1, and added R2 exists after B1. The complete no-add stories are RR,RB,BR,BB, all equally likely, giving one of each in 2/4=1/2. Fractions can be compared using six and four equally weighted models or an adult-supported common denominator; raw successes 4,2,2 alone are not a chance proof.

With opposite reinforcement final bags and draw totals reverse the usual copying correspondence: RR ends 1R/3B, RB/BR end 2R/2B, BB ends 3R/1B. All original colors persist under every rule. Hints: after the first R, inspect which identities are available next; keep the two additions visibly different. Extension: compare three draws and ask whether the opposite rule always favors the most balanced totals; do not claim a general law from two draws.

New versus base: base middle/upper Problem 3/2 compares copying with no additions. The new mechanism is opposite-color reinforcement and its balance-promoting distribution. The familiar two controls support the new comparison and are not separately counted as investigations.

Problem 2 (page 2)

Strongest red: RRRR, final 5R/1B, next red 5/6. Weakest red: BBBB, final 1R/5B, next red 1/6. Tied: any of RRBB,RBRB,RBBR,BRRB,BRBR,BBRR, final 3R/3B, next red 1/2. None can forecast certainty because both original colors persist.

Other attainable forecasts are 4R/2B (2/3) and 2R/4B (1/3). For example RRRB and RRBR give the first, RBBB and BRBB the second. The extreme forecast has only one four-draw color story; every intermediate forecast has multiple orders. Children may give “five of the six counters” rather than symbolic fractions. A live partner bet can test the executable forecast but cannot establish its exact probability through a small sample.

Hint: put the current counters in a visible row and ask how many would win the next red bet. Extension: after a red next draw, update a formerly tied bag and its forecast; show that later evidence changes the prediction again. These are continuations within the forecasting investigation.

New versus base: the base enumerates finishing bags and equal-count story chances. This reverses the direction to design a predictive bag and forecast the NEXT random draw conditional on the current state, with extremal and tied targets.

Problem 3 (page 3)

The twelve printed stories are all original identities available after the first draw plus that draw's one new copy. Each has probability 1/(3×4). The six final bags, in R/B/G order, are (3,1,1),(1,3,1),(1,1,3),(2,2,1),(2,1,2),(1,2,2). Each gets two identity histories. A repeated word such as RR has two histories (R1,R1),(R1,R2); a mixed word RG has one, but its reversed word GR adds the second history to the same final bag.

After three draws, added vectors are permutations of (3,0,0), of (2,1,0), and (1,1,1): three plus six plus one = ten. Their final counts add one to every coordinate. Each vector has six of the sixty marked histories. For (3,0,0), the one color word has 1×2×3=6 identity histories; for a (2,1,0) vector, three word orders each have 2!=2 histories; for (1,1,1), six distinct orders each have one history. Those three types cover all nonnegative triples summing to three.

General adult explanation: for fixed vector (r,b,g), each color word has the same product r!b!g! of available-color choices. There are n!/(r!b!g!) words, so multiplying gives n! marked histories per vector. Total histories are 3×4×...×(n+2)=(n+2)!/2, yielding 1/C(n+2,2). Children can work with paired rearrangements and grouped small cases; factorial notation is not required.

Hints: ignore identity labels only AFTER reconstructing the final bag; keep a counter of each original color in the tray record. Extension: arrange the vectors as points in a triangular count board; ask what neighboring points mean when one added color is changed. This is an optional next representation, not required alongside concurrent draw tallies.

New versus base: the base proves the binary total law, including arbitrary n. Three colors introduce a two-dimensional count vector, triangular number of states and equal weight across count bags despite unequal ordered-word weights.

Sources, checks, and actual-use record

Original extensions of current Week 44 copying materials. The base cites Janko Gravner, UC Davis MAT 235B, Martingale convergence theorems (February 2, 2024), slides 12-14, Pólya urn. The finite three-color proof and changed-feedback/forecasting tasks here are independently derived; no martingale prerequisite is imposed. Pedagogy: Math Circle by the Bay, Preface pp. viii-x (PDF pp. 9-11), deep themes, manipulatives and variable duration. These particular returns are design inferences. The Week 1 encore and old extension model were consulted.

student/verify.py separately enumerates every marked two-draw history under each rule, all sixteen four-draw color stories for predictions, all printed three-color history cards, and three-color laws through five draws. The independent mathematics review confirmed all histories and laws through five draws. All three final student pages were rendered and individually inspected after revision. Portable verifier and extracted-source rebuild pass; record preparation is specified in the guide and remains physically untested. **Unpiloted; the banker/reset procedure, physical counter identities, cutting and mixing have not been pretested.**

Record date/adult, selected investigation, rule retained independently, child's current-bag forecasts or sorted cases, experiment/conjecture/explanation status, confusion and next starting point. One theme, three investigations; no actual use yet claimed.